

KAPIL

Dehradun, Uttarakhand, India

+91-9996547555 ky8050989@gmail.com [Portfolio](#) [LinkedIn](#) [Github](#)

EDUCATION

University of Petroleum and Energy Studies

B.Tech in Computer Science and Engineering - CGPA: 7.58/10

2023 - 2027

Dehradun, Uttarakhand

EXPERIENCE

IBM – Software Engineering Intern

Remote June 2026 - July 2026

- Built an AI due-diligence platform using LLMs and vector databases (FAISS), automating analysis of legal, financial, and compliance documents; reduced manual review effort by ~60%.
- Designed and shipped a multi-agent workflow for contract review, financial risk scoring, and automated report generation, reducing risk identification time by ~40%.
- Shipped production features end-to-end from system design through implementation in a cross-functional team; incorporated iterative feedback to deliver on schedule.

Weazy – Web Developer Intern

Remote June 2025 - Aug 2025

- Engineered a full-stack Flask REST API automating product image fetching, processing, and database updates for an e-commerce catalog; eliminated manual effort across 10K+ SKUs.
- Built intelligent web-scraping pipelines with Selenium to fetch relevant images from Google, Bing, and DuckDuckGo with filtering, preview, and human-in-the-loop selection logic.
- Deployed a cloud-ready microservices solution on Azure with modular design and CI/CD pipeline (GitHub Actions); wrote clear technical documentation and contributed to cross-team design discussions.

PROJECTS

Corex

LangChain, FastAPI, Docker, OCR, FAISS

- Built a retrieval-first RAG assistant with FastAPI and FAISS; implemented a multi-stage fallback pipeline with relevance-threshold filtering to suppress low-confidence matches, serving grounded, source-cited answers.
- Reduced average end-to-end retrieval-to-generation latency to under 2 seconds via NVIDIA-hosted embedding caching and FAISS index tuning.
- Built multimodal document ingestion (PDF, image via OCR) alongside browser-native voice input and text-to-speech; deployed via Docker on zero-GPU cloud hosting.

AETHER

API Gateway, FastAPI, SSE, Circuit Breaker, Docker

- Built a Python LLM API gateway with SSE streaming; implemented multi-backend routing (Ollama default, HuggingFace fallback), a minimal local web UI, and containerized Docker/Kubernetes deployment.
- Engineered production-grade resilience: circuit breaker, exponential backoff retry, and token-bucket rate limiting to maximize inference throughput and availability.
- Integrated API-key and JWT authentication with Prometheus metrics and structured logging demonstrating end-to-end observability.

NovelNet

Django, Supabase, HTML, Google Books API

- Designed and deployed a full-stack Django book review and recommendation platform serving 100+ users, integrating Google Books API, JWT authentication, and cloud file management.
- Built secure file handling with user attribution and REST API endpoints via Django REST Framework for mobile client integration.
- Deployed on Gunicorn and PostgreSQL; configured scalable DB connection pooling and secure session management.

TECHNICAL SKILLS

Languages: Python, C++

Backend & Systems: FastAPI, Django, Flask, gRPC, SSE, JWT/OAuth 2.0, REST API

ML/AI: TensorFlow, PyTorch, scikit-learn, OpenCV, LangChain, RAG, Vector Databases, FAISS

System Design: Data Structures & Algorithms, DBMS, Operating Systems, Microservices, API Design

DevOps: Docker, Kubernetes, CI/CD (GitHub Actions, GitLab, Jenkins), Azure VM, Azure ML

Databases: PostgreSQL, MySQL, Redis

ACCOMPLISHMENTS & CERTIFICATIONS

- Advanced to the interview stage at HackWithInfy, clearing Coding Rounds I and II.
- Reached Final Round of Deloitte Hackspllosion ranked among Top 60 teams nationwide.
- Pre-trained a GPT-style 13.6M parameter model from scratch (custom tokenizer, transformer architecture).
- Solved 250+ Problems on LeetCode
- Oracle Data Platform 2025 Certified Foundations Associate – Oracle